

Taylor series - 6.3

Knowing that $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$ on $(-1, 1)$ has been enough to find series for $\frac{1}{1+x}$, $\frac{1}{1+x^2}$, $\arctan(x)$, etc

But what about other functions, say, $f(x) = e^x$?

Let's just suppose e^x does have a power series representation, say
$$e^x = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots$$

Try to figure out coefficients by plugging in some values. Only know this should converge

at $x=0$.
$$e^0 = c_0 + c_1(0) + c_2(0)^2 + \dots = c_0,$$

so we see that $c_0 = 1$.

$e^x = \sum_{n=0}^{\infty} c_n x^n$ has some radius convergence R ,

and we know that $e^x = \frac{d}{dx} e^x = \sum_{n=0}^{\infty} \frac{d}{dx} (c_n x^n)$

for x within R of 0.

$$e^x = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots$$

$$\downarrow \frac{d}{dx}$$

$$e^x = \sum_{n=1}^{\infty} n c_n x^{n-1} = 0 + c_1 + 2c_2 x + 3c_3 x^2 + \dots$$

Plugging in $x=0$ gives $1 = e^0 = c_1 + 2c_2(0) + 3c_3(0)^2 + \dots = c_1$

So we now know $c_0 = 1$ and $c_1 = 1$.

Differentiating again:

$$e^x = \sum_{n=2}^{\infty} (n-1)n c_n x^{n-2} = 2c_2 + 3 \cdot 2c_3 x + 4 \cdot 3c_4 x^2 + 5 \cdot 4c_5 x^3 + \dots$$

at $x=0$ we get $1 = e^0 = 2c_2 \Leftrightarrow c_2 = \frac{1}{2}$.

Again!

$$e^x = \sum_{n=3}^{\infty} n(n-1)(n-2) c_n x^{n-3} = 3 \cdot 2 c_3 + 4 \cdot 3 \cdot 2 c_4 x + 5 \cdot 4 \cdot 3 c_5 x^2 + \dots$$

at $x=0$ we get $1 = e^0 = 3 \cdot 2 c_3 \Leftrightarrow c_3 = \frac{1}{3 \cdot 2}$

If we take k derivatives:

$$e^x = \sum_{n=k}^{\infty} n(n-1)(n-2) \dots (n-k+1) c_n x^{n-k}$$

plugging in $x=0$

gives $e^0 = \text{constant term.}$

$$e^x = \sum_{n=k}^{\infty} n(n-1)(n-2)\dots(n-k+1)c_n x^{n-k} \quad \text{at } x=0$$

$$\begin{aligned} 1 = e^0 &= \text{constant term } n=k \\ &= k(k-1)(k-2)\dots(k-k+1)c_k x^{k-k} \\ &= k(k-1)(k-2)\dots 1 \cdot c_k = k! \cdot c_k \end{aligned}$$

Rearranging gives $c_k = \frac{1}{k!}$.

Therefore, if e^x has a power series representation,
it must be $e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$.

Generalizing this process, we take some function $f(x)$ and some number a .

If we can write $f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n$,

then taking k derivatives gives

$$f^{(k)}(x) = \sum_{n=k}^{\infty} n(n-1)(n-2)\dots(n-k+1) c_n (x-a)^{n-k}$$

At $x=a$ this says $f^{(k)}(a) = k(k-1)(k-2)\dots 1 \cdot c_k = k! c_k$

and therefore $c_k = \frac{f^{(k)}(a)}{k!}$.

Theorem: If $f(x)$ has a power series representation centered at a , i.e. if $f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n$, then $f^{(n)}(a)$ exists for all n and $c_n = \frac{f^{(n)}(a)}{n!}$.

This doesn't guarantee such a power series is actually equal to $f(x)$, but we can still write down the series.

Definition: If $f(x)$ has all its derivatives at a , then its Taylor series is the power

series $\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$. (If $a=0$, sometimes

we call this series the Maclaurin series.)

